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# Unveiling the Complexity of Designers' Intention to Use Generative AI in Corporate Product Design: A Grounded Theory and fsQCA

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**Abstract:** While generative artificial intelligence (Gen AI) is accelerating digital transformation and innovation in corporate product design (CPD), limited research has explored how designers adopt this technology. This study aims to identify the key factors and causal configurations that influence designers' intentions to adopt Gen AI in CPD. This study involved 327 in-service designers as participants, employed semi-structured interviews and a questionnaire to collect data, and applied the grounded theory and fsQCA to analyze the data. The findings indicate the following: (1) Personal innovativeness, AI technological anxiety, perceived usefulness, task–technology fit, perceived risk, social influence, and organizational support are the key factors influencing designers' adoption of Gen AI. (2) None of these factors constitute a necessary condition for designers to adopt Gen AI. (3) High adoption intention results from the interaction of multiple factors, which can be categorized into three driving logics: "task demand-driven", "organizational environment-driven", and "individual characteristics-driven". It is recommended that corporate managers establish an AI training framework, foster a supportive organizational environment, and implement tailored strategies to facilitate the integration of new technologies. This study clarifies the factors influencing designers' adoption of Gen AI in CPD and provides a framework for companies to effectively integrate AI systems into product design.

**Keywords:** corporate product design; generative AI; adoption intention; grounded theory; fsQCA

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## 1. Introduction

Generative AI (Gen AI) is an artificial intelligence technology capable of creating various forms of content in response to user prompts such as text, images, videos, code, and design solutions [1,2]. In recent years, corporate product design (CPD) teams have increasingly adopted domain-specific Gen AI tools like Midjourney, Dreamcatcher, and Magic3D. These Gen AI systems surpass traditional computer-aided design tools by acting as "co-creators" [3]. They provide tangible benefits across CPD stages, including problem analysis [4], ideation [5], solution evaluation [6], and product development [7]. A global study by McKinsey in August 2023 (covering over 200 companies across medical products, automotive, and electronics manufacturing, with 1684 participants) revealed that 27% of companies using Gen AI in product design experienced a 6% to 10% increase in sales revenue due to innovation [8]. Similarly, recent studies in China indicated that

Gen AI showed significant potential in improving product design efficiency and quality, including a 50% reduction in development time, a 60% acceleration in innovation, and the creation of groundbreaking products [9–11]. Overall, Gen AI has transformed traditional corporate product design approaches to improve design efficiency and quality [12]. This offers possibilities for addressing the limitations of traditional labor-intensive designs and facilitating digital transformation.

Despite its potential advantages, integrating Gen AI into CPD presents numerous challenges. Research indicates that at the beginning of 2023, only 13% of enterprises incorporated Gen AI into CPD. As of April 2024, this figure remains low, with only 18% of enterprises in the United States, 31% in Germany, and 37% in China adopting Gen AI [13]. At present, many companies are still in the early stages of gathering information regarding generative AI and have yet to reach the expected level of adoption [12,14]. Additionally, due to designers' insufficient AI literacy [15], skepticism [16], risk assessment concerns, and past failure experiences [3], their intention to adopt Gen AI remains weak, with some even refusing to use new technologies. This phenomenon weakens enterprises' ability to innovate and stay competitive while also hindering CPD's digital transformation and sustainable development. Since adoption intention is the strongest predictor of adoption behavior [17], it is important to understand the factors that influence designers' intention to adopt Gen AI and to promote its effective use to maximize its benefits.

The existing research has explored various factors influencing the adoption of Gen AI in the design field. They include studies on how design students use Gen AI in coursework [16], the adoption behavior of creative industry professionals [5,18], and the factors influencing individual designers' adoption intentions [19,20]. However, these studies mainly examined adoption at the individual level and did not fully address decision-making in specific work settings. Research on designers' adoption of Gen AI for specific tasks remains limited. Among the few available studies, Xia identified performance expectancy, anthropomorphism, and task compatibility as key factors influencing designers' adoption of Gen AI in new product development [21]. However, their research primarily focused on individual cognitive and behavioral characteristics, overlooking the challenges within organizational contexts. Compared to general contexts, Gen AI adoption in CPD is more complex, as it is influenced by technical, organizational, and environmental factors [22,23] while also being constrained by corporate resources, infrastructure, and system compatibility [24,25]. Moreover, previous studies have relied on single-variable analysis methods, such as regression analysis or structural equation modeling (SEM), to assess the net effect of a single factor, which limits the understanding of the complex interactions between multiple factors [26].

Therefore, this study employed a two-stage approach. First, grounded theory was applied to construct a comprehensive model. Then, we utilize fsQCA to examine the factors shaping designers' adoption of Gen AI in CPD and their complex interrelationships. Specifically, this study aimed to address the following research questions:

RQ1: Across the entire product design cycle, from problem definition to product development, what factors influence designers' intention to adopt Gen AI?

RQ2: How do these factors interactively combine to influence designers' adoption intention?

This study makes several important contributions to theory, methodology, and practice. Theoretically, our comprehensive model, developed through grounded theory, advances the understanding of designers' Gen AI adoption mechanisms in CPD, thereby extending existing theoretical frameworks. Methodologically, this study is among the first to utilize fsQCA, an asymmetric approach, to uncover the ways in which various factors can be combined to stimulate designers' intention to use Gen AI. This novel analytical technique

serves as an alternative and complementary method to traditional symmetric approaches (e.g., regression and structural equation modeling) that have been widely used in prior research to analyze the "net effect" of antecedents on Gen AI adoption. Focusing on net effects without considering the complex configurations among variables can lead to erroneous results [27]. Our fsQCA results revealed five distinct and equally effective pathways leading to designers' adoption of Gen AI in CPD. From a practical perspective, our findings provide guidance for enterprise managers to promote the widespread adoption of Gen AI in CPD, contributing to digital transformation and sustainable development.

## 2. Literature Review

### 2.1. Gen AI in the Corporate Product Design Process

Corporate product design (CPD) is a multi-stage systematic process that includes problem definition, business analysis, idea generation, testing and evaluation, and product development [28–30]. Each stage has distinct task characteristics: the early stages of problem definition and business analysis focus on information gathering and data analysis; the idea generation stage emphasizes diverse creative concepts; the testing and evaluation stage primarily involves screening and optimizing solutions; and the development stage covers specific engineering tasks such as product modeling, manufacturing, and programming [21].

In the problem definition and business analysis stage, Gen AI reshapes traditional methods of market and user research. Traditional approaches involve manual gathering and analysis of user and market data [31]. In contrast, Gen AI tools, including Monday, Funnel, Azure, and ChatGPT, facilitate more efficient user research and market analysis by rapidly processing and analyzing vast amounts of unstructured data to identify pain points and emerging product demands [12,32]. For example, Nestlé collaborated with CI&T to leverage an AI platform equipped with NLP and image recognition technologies to scan online data sources and extract user needs, thereby generating novel packaging design concepts and increasing innovation efficiency [33]. Similarly, Zalando employed Project Muze to continuously gather and analyze customer feedback and suggestions. By leveraging trend image analysis to identify popular styles and market trends, Zalando enhanced design effectiveness and efficiency, supported early-stage decision-making, reduced design and development time, and lowered costs [34]. Furthermore, Gen AI can analyze customer feedback and sentiment, converting complex data into actionable insights that support decision-making, reduce risks, and improve the efficiency of design analysis [4,35].

The ideation stage aims to generate numerous high-quality conceptual solutions within a limited timeframe. Designers have traditionally expanded their creative horizons by gathering external stimuli, primarily through mood boards [36]. Furthermore, emerging AI tools, including DALL·E, Artbreeder, Stable Diffusion, and GPT, allow designers to generate creative ideas quickly and cost-effectively, thereby providing abundant sources of inspiration [37,38]. For instance, IDEO employed GPT to facilitate a brainstorming session to develop a new banking application that generated five promising conceptual solutions that address different user needs [39]. Moreover, AI-generated concepts serve as preliminary drafts, facilitating cross-functional reviews among design, development, and testing teams. These collaborative reviews enable team members to discuss design drafts, identify potential issues, and promptly propose optimization strategies, significantly reducing feedback cycles while enhancing cross-team communication efficiency [40].

During the testing and evaluation stage, designers must assess numerous design solutions and select the most promising candidates for further development [41]. Traditional methods predominantly rely on expert evaluation, which poses significant challenges, including high costs, time consumption, and inconsistencies owing to subjective interpre-

tations [42,43]. Recent advancements in Gen AI have enabled the quantitative analysis of design elements, such as color schemes and layout structures, while incorporating online trend analysis to deliver objective assessments closely aligned with human perception [44]. These systems have demonstrated remarkable efficacy in evaluating UI design consistency, offering applications both within single platforms and across multiple interfaces [45]. Moreover, they facilitate multidimensional data analysis and simulation testing, thereby allowing rigorous design evaluation. A notable example is Stryker's use of AI-driven analysis to evaluate the compatibility between novel medical devices and human anatomical structures, successfully identifying design limitations before manufacturing [46].

In the product development phase, Gen AI accelerates the design and manufacturing process by automating rendering, digital modeling, and programming tasks. These systems handle repetitive and technically demanding operations, such as creating 3D models [47], generating design renderings [48], and optimizing designer proposals [49]. Consequently, designers can focus on the creative and strategic aspects of their work. Tools such as Midjourney and Stable Diffusion generate design renderings based on designer prompts, significantly reducing the time traditionally spent on drawing, modeling, and rendering [50]. In addition, advanced AI models provide programming support through automatic code completion [51,52], natural language-to-code conversion [53], and duplicate code detection [54]. For example, Hikvision leveraged large language models from Anthropic and OpenAI for code generation, thereby achieving substantial improvements in programming efficiency [55].

The full realization of Gen AI's advantages depends heavily on designers' intentions to accept and adopt the technology. Close collaboration between designers and Gen AI is essential for unlocking its full potential. This raises a critical question: what factors influence designers' adoption of Gen AI in CPD? Despite its significance, this topic remains underexplored in the field of design.

### *2.2. Theoretical Models and Influencing Factors of Gen AI Designers' Adoption Intentions*

Previous studies have employed various theoretical models to explore the factors influencing designers' adoption of Gen AI. Commonly adopted theoretical frameworks include the Technology Acceptance Model (TAM) [56], the Unified Theory of Acceptance and Use of Technology (UTAUT) [57], its extended version UTAUT2 [58], the Theory of Planned Behavior (TPB) [59,60], Social Cognitive Theory (SCT) [61], and the Expectation-Confirmation Model (ECM) [62]. Many studies have employed structural equation modeling (SEM) to develop conceptual models. Du et al. integrated AI anxiety and AI literacy into the UTAUT framework to elucidate designers' attitudes and behavioral intentions toward AI-powered drawing tools [63]. Xia et al. investigated the determinants of Gen AI adoption and designers' attitudes toward new product design tasks by leveraging both UTAUT and UTAUT2 [20]. Fan et al. combined ECM with the Information Systems Continuance (ISC) model to analyze designers' intentions toward adopting AI-based drawing tools [64]. Additionally, Jiao et al. synthesized TPB and TAM to investigate designers' adoption of AI-assisted design tools [19]. Although these models have widespread applicability in explaining individual technology adoption behaviors, they primarily focus on individual-level cognitive and behavioral decision-making. However, CPD involves complex factors, such as team collaboration, organizational culture, and management decisions. Therefore, relying solely on these models may not fully capture the dynamic interactions and organizational factors that designers face in the AI adoption process.

The core predictors within UTAUT2 play a crucial role in shaping designers' intentions to adopt Gen AI. Empirical studies have shown that perceived usefulness, perceived ease of use, and social influence positively affect the adoption of Gen AI [21,63]. In addition

to these dimensions, factors derived from fundamental theoretical models have attracted considerable research interest. Wang et al. found that subjective knowledge and perceived enjoyment strongly affect Chinese design students' intentions to use AI-based design tools [11]. Similarly, Li found that perceived anxiety and perceived risk negatively impact both design students' and professional designers' intentions to adopt Gen AI [20]. However, these studies focused on the impact of individual psychological and cognitive factors, such as "perceived ease of use" and "perceived usefulness", on adoption decisions, which do not fully address decision-making issues in specific work environments. In the CPD context, designers' decisions to adopt Gen AI are often influenced by the combined effects of technological, organizational, and environmental factors [22,23] and are constrained by variables such as organizational culture, resources, and system compatibility [24,25].

Moreover, these studies often use quantitative methods, such as SEM and theoretical frameworks, to incorporate or exclude various variables to refine them. This approach may overlook important influencing factors or result in overly complex models that lack practical relevance. Additionally, the factors influencing designers' adoption of Gen AI are interconnected and interdependent, and this complex and asymmetrical causal relationship requires more suitable analytical methods [26,65].

Given these concerns, this study employed an integrated methodology that combines grounded theory and fsQCA. Grounded theory, through in-depth interviews, uncovers key factors influencing corporate designers' adoption of generative AI in the CPD context, particularly those overlooked by predefined theoretical frameworks [66]. Building on these insights, a comprehensive model is developed to consolidate the determinants shaping adoption intentions. Subsequently, fsQCA is applied to identify multiple configurations and alternative pathways that lead to a high willingness to adopt Gen AI, thereby enhancing both the specificity and generalizability of the findings [67,68]. This methodological integration not only addresses the limitations of existing research but also provides actionable insights for enterprises to optimize AI adoption and innovation strategies.

## 3. Methodology

This study employs an exploratory sequential mixed methods design to investigate factors influencing designers' adoption of Gen AI in CPD [58]. By integrating qualitative and quantitative data, this approach offers a comprehensive understanding of the phenomenon [69,70] and strengthens the rigor and validity of the findings through methodological triangulation [71,72]. To capture the complexity of designers' adoption intentions, this study combines grounded theory with fsQCA, enabling an exploration of not only individual influencing factors but also their configurational interrelationships. The research comprised two phases: Study 1 and Study 2 (see Figure 1).

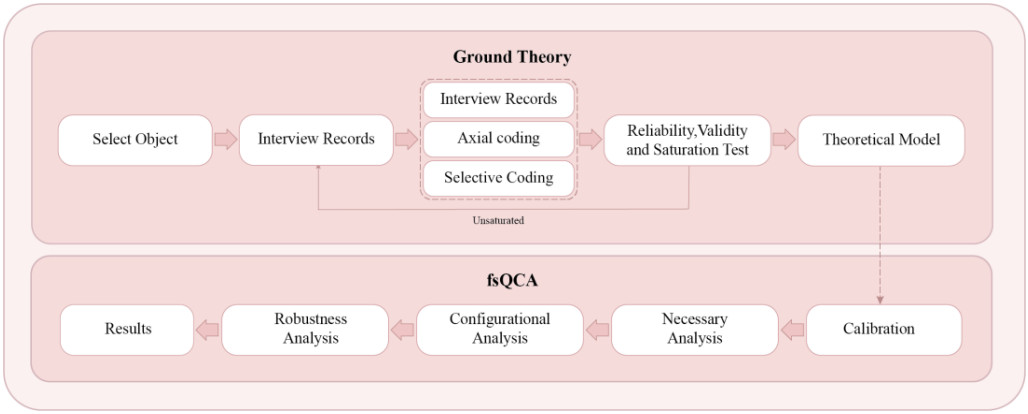



**Figure 1.** Research methods and structure.

Study 1 involved semi-structured interviews with 45 designers from various Chinese enterprises. Using grounded theory, this study identified key factors shaping designers' intentions to adopt Gen AI and revealed the relationships among them.

Grounded theory requires researchers to deeply investigate the context, grounding their research in real-world data rather than simply extracting insights from the existing literature. This enables the integration of new observations with established theories to provide a more comprehensive explanation of specific phenomena [27]. While variables in models such as TAM and UTAUT (e.g., performance expectancy, effort expectancy, and social influence) are somewhat related to designers' intention to adopt Gen AI, the extent to which these factors influence CPD designers' adoption intentions and whether other factors also affect this intention have not been fully explored in the current research. This gap calls for further investigation and validation. Moreover, these models primarily address general technology adoption contexts and fail to fully capture the unique factors that designers face in product design scenarios, such as individual characteristics and professional influence. Grounded theory, through systematic data collection and analysis, can inductively develop a theoretical framework that better explains the essence of a phenomenon, making it particularly suitable for constructing theories around underexplored phenomena or those insufficiently explained by existing theories [73,74]. Therefore, this study adopts the classic grounded theory approach, consisting of open coding, selective coding, and theoretical coding, to systematically induct and refine the factors influencing designers' intention to adopt Gen AI in CPD.

In Study 2, a questionnaire and fsQCA were used to examine the relationship between designers' adoption intentions toward Gen AI and the influencing factors. The use of fsQCA is justified for three main reasons. First, designers' adoption intentions are shaped by multiple interacting factors, and fsQCA effectively captures such complex configurations [75]. (2) Second, it identifies combinations of conditions that lead to adoption, offering insight into potential causal mechanisms. Third, fsQCA accounts for asymmetry by analyzing both the presence and absence of conditions, enabling a more nuanced understanding of adoption pathways [27]. Despite its advantages, fsQCA has limitations, such as sensitivity to variable selection and potential subjectivity in result interpretation. To address these concerns, variables were derived from the grounded theory findings in Study 1, ensuring a strong theoretical foundation. A transparent calibration process and sensitivity analyses were conducted to ensure robustness. Finally, the results were interpreted in conjunction with qualitative insights to reduce subjectivity and strengthen the validity of the findings.

In summary, this study employs a mixed methods design to examine the factors influencing designers' intentions to adopt Gen AI in the CPD context. The qualitative phase identifies key factors, while the quantitative phase, using fsQCA, validates these findings and explores their interrelationships. The methodologies, data collection processes, and results of Studies 1 and 2 are detailed in the following sections.

## 4. Study 1: Adoption Intention and Its Influencing Factors: A Grounded Theory Approach

### *4.1. Interview Design*

This study employed individual interviews, ensuring participant privacy and eliciting authentic insights, attitudes, and personal experiences, leading to a comprehensive understanding of the issues [76]. A semi-structured interview framework was adopted to balance topic control with participant autonomy. Regarding the design of the interview outline, the research team first developed an initial draft based on a review of the relevant literature and practical observations. Next, on 13 October 2024, a focus group discussion was conducted with three designers experienced in using Gen AI tools, including ChatGPT,

Midjourney, and Magic3D, to refine certain questions based on their feedback. Finally, four experts with extensive experience in product design and user behavior research were invited to discuss relevant topics, review the interview outline, and implement necessary revisions, resulting in a finalized open-ended interview outline. The interview questions included those listed in Table 1, but were not limited to these. These questions provided an open-ended framework, allowing for follow-up inquiries based on the respondents' answers, thereby enabling the researcher to probe deeper into potential new factors.

**Table 1.** Interview outline.

| Number | Question |
| --- | --- |
| 1 | Which Gen AI tools have you used to assist with design work? How have they improved your work efficiency or design quality? |
| 2 | Compared to traditional design methods, what do you consider to be the main advantages and limitations of Gen AI? |
| 3 | What are the differences in applicability of different types of Gen AI in design work? Which specific functional tools are you more inclined to choose? |
| 4 | When using Gen AI in your work, do you have any concerns related to data privacy and security? |
| 5 | Have you ever changed your attitude toward Gen AI due to the perspectives of your colleagues or industry peers? |
| 6 | Does your company provide support for using Gen AI (such as technical resources, funding, or training)? How has this support helped you? |
| 7 | Will you continue to use Gen AI in the future? What are your suggestions or expectations? |

### *4.2. Sample Selection*

This study employed purposive sampling to recruit participants with experience using Gen AI tools, ensuring relevance to the research theme [77]. The selection criteria were as follows: (1) participants must be currently employed designers who have used Gen AI tools in their design work (e.g., ChatGPT, Midjourney, Stable Diffusion); (2) the sample should include diverse age groups, work experiences, professional domains, and industries to minimize biases and enhance the generalizability and reliability of the research conclusions.

Between 2 and 13 October 2024, through the first author's design project, we initially established contact with Hikvision. Subsequently, we invited the project leader to act as an intermediary and used a snowball sampling strategy to extend the study to companies from different regions and industries [78]. Research indicates that medium and large enterprises are the primary early adopters of Gen AI [9], and such organizations typically have more standardized design functions and processes, which is conducive to focusing on our research questions. Furthermore, during the investigation, we found that the design functions in some small enterprises were relatively unclear, and because of resource limitations and business operation characteristics, many designers have not widely applied Gen AI in their design work. Therefore, we ultimately decided to exclude small enterprises and focus on medium and large enterprises to address our research questions. Through this sampling process, we successfully invited 45 professional designers from various industry fields to participate in this study (Z1–Z45).

Regarding demographic characteristics, 47.8% of the sample were male and 52.2% were female. From an enterprise perspective, designers are distributed across various industries, including automotive (e.g., Chery and Volkswagen), medical (e.g., Mindray), security (e.g., Hikvision and Dahua), Internet (e.g., NetEase and Tencent), finance (e.g., HTSC), education (e.g., iFlytek and Seewo), equipment manufacturing (e.g., XCMG), and consumer electronics (e.g., Xiaomi and Huawei). The participants' professional experience varied across career stages: 42.2% had 0–3 years of experience, 36.7% had 3–5 years, 14.4% had 5–10 years, and 6.7% had more than 10 years of experience. In terms of industry roles,

the majority were product and industrial designers (64.8%), followed by UX/UI designers (29.4%), while the remaining 5.8% were user researchers, product managers, and team leaders (see Table 2).

**Table 2.** Participants' demographic information.

| Item | Indicator | Frequency | Percentage (%) |
|---|---|---|---|
| Gender | Male | 156 | 47.8 |
| | Female | 171 | 52.2 |
| Age (years) | 18–25 years | 109 | 33.3 |
| | 25–35 years | 153 | 46.8 |
| | Over 35 years | 65 | 19.9 |
| Years of experience (years) | 0–3 years | 138 | 42.2 |
| | 3–5 years | 120 | 36.7 |
| | 5–10 years | 47 | 14.4 |
| | Over 10 years | 22 | 6.7 |
| Role | PD | 125 | 38.2 |
| | ID | 87 | 26.6 |
| | UX/UI | 96 | 29.4 |
| | Others | 19 | 5.8 |
| Field | Automotive | 47 | 14.4 |
| | Medical | 34 | 10.4 |
| | Internet | 44 | 13.5 |
| | Education | 38 | 11.6 |
| | Finance | 32 | 9.8 |
| | Security | 54 | 16.5 |
| | Equipment | 41 | 12.5 |
| | Others | 37 | 11.3 |

### *4.3. Data Collection*

The research team conducted interviews with participants from 15 October to 12 November 2024, collecting detailed data, including notes and recordings. Interviews with designers in East China were held face-to-face in corporate meeting rooms, while video or telephone interviews were conducted for participants in other locations. Previous qualitative research has supported the validity of telephonic interviews. Compared to face-to-face interactions, telephone interviews provided participants with a greater sense of control [79]. Moreover, studies have shown that telephone interviews can yield equally reliable results, particularly when the focus is on response content rather than interactive dynamics [80,81]. These findings reinforce our confidence in using video and telephone interviews as an effective data-collection method.

Before the interviews, the researchers explained the purpose and content of this study to the participants and obtained authorization for audio recording and documentation. To ensure the protection of participants' rights, all personal information and data were anonymized, and participants were informed of their right to withdraw from the study at any time. This study complied with the ethical principles outlined in the Declaration of Helsinki. This study was approved by the Academic Ethics Committee of Wuhan University of Technology's College of Arts in September 2024 (approval number: 20241117).

The interviews lasted 30 to 50 min, and all recordings were transcribed verbatim and cross-verified immediately. Contact information was collected for potential follow-up interviews and data verification. All textual data were managed and coded using NVivo 12.0, following grounded theory principles. The coding process consisted of three phases: open, axial, and selective coding [82]. Stratified sampling was applied, with two-thirds (30 partic-

ipants) used for detailed analysis and the remaining one-third (15 participants) used for theoretical saturation verification. Data extraction, induction, and analysis were performed iteratively until saturation was reached.

### 4.4. Data Analysis

#### 4.4.1. Open Coding

Open coding is the process of conceptualizing and categorizing data derived from interview transcripts. This process involves deconstructing, reorganizing, and assigning conceptual labels to the original text, thereby identifying the key factors influencing the adoption intentions of Gen AI designers [83]. To ensure objectivity and scientific rigor, two coding teams were formed, each led by a senior expert to enhance the analytical rigor of the process. A systematic coordination mechanism was established between the teams. The team leaders convened weekly meetings where both groups conducted comprehensive analyses and thorough discussions of the coding outcomes to ensure consistency and reliability.

The research team employed NVivo 12 to perform a comprehensive sentence-by-sentence analysis of the raw textual data by removing ambiguous or irrelevant statements. This process yields 976 initial sentence tags. Through systematic consolidation and conceptual analysis, 79 distinct concepts were identified. Given the large number of concepts and overlapping contents, further classification is required. Concepts were categorized according to their causal relationships, similarities, and relational types. Considering the extensive sample size, concepts with low-occurrence frequencies ($\leq$2 times) were excluded to maintain analytical focus on more frequently observed concepts. Ultimately, 21 subcategories were identified and labeled as A01 through A21 (see Table 3).

**Table 3.** Open coding analysis results.

| Subcategory | Initial Category (Partial) | Original Sentence (Partial) |
| --- | --- | --- |
| A01 Proactivity and Enthusiasm | Initiative-taking behavior<br>Experiential engagement<br>Learning enthusiasm | Whenever SD updates its features, I always try them out immediately (Z21).<br>I find the process of exploring AI quite interesting, even if the design outcomes are not always ideal (Z22).<br>I think personal character may have some influence. For example, some colleagues seem to just work for the paycheck and tend to stay within familiar areas. As for me, I'm relatively more proactive and still have some enthusiasm for doing these things (Z27). |
| A02 Exploratory Motivation | Sustained exploration<br>Learning and adaptation | I often try different prompts to explore what kind of unique results AI can bring me (Z23).<br>Maybe my "magic words(prompt)" weren't good enough, and when I first started using AI, the results weren't ideal. However, after continuous adjustments, I achieved satisfactory results (Z01). |
| A03 Creative Autonomy | Innovation process<br>Decision-making leadership | I am willing to adjust my original design process and integrate AI into it (Z05).<br>I will adjust AI's output based on my own design approach (Z01). |
| A04 Skill-related Anxiety | Design competency anxiety<br>Creative ability anxiety | I think the illustrations created by MJ are better than the ones I drew (Z08).<br>The involvement of AI makes me feel that my creative space has shrunk, as I tend to rely on the machine rather than my own ideas (Z21). |

**Table 3.** *Cont.*

| Subcategory | Initial Category (Partial) | Original Sentence (Partial) |
| --- | --- | --- |
| A05 Low Self-efficacy | Learning helplessness<br>Lack of perceived control<br>Self-doubt | AI tools are updating so quickly that I don't even know where to start learning (Z12).<br>The design results generated by AI often deviate from my original intentions, and I feel like it's hard to fully control the design process (Z26).<br>I feel that my prompt skills are too weak, and the results generated by AI are always unsatisfactory (Z18). |
| A06 Career Development Anxiety | Career uncertainty<br>Professional identity crisis | I'm worried that companies will rely more on AI in the future, reducing the demand for designers, or even no longer needing us (Z09).<br>Transitioning from a designer to an AI operator, I feel that my professional value has been diminished (Z11). |
| A07 Work Efficiency Enhancement | Design task automation<br>Assisted creative generation<br>Information processing optimization | I think using Magic3D to generate models is much faster than using Rhino, and it can generate many alternative options (Z03).<br>With AI handling the initial conceptualization, the design process becomes faster and easier (Z05).<br>In the past, understanding requirements took a lot of time, but now GPT can directly help me analyze them (Z17). |
| A08 Design Quality Enhancement | Precise data analytics<br>Improved design outcomes<br>Personalization adaptation<br>Creative expansion | AI's suggestions for user demand analysis and competitive product analysis are very practical and reliable, in my opinion (Z27).<br>The renderings generated by AI are stunning, with precise attention to detail (Z18).<br>AI can quickly produce customized designs that meet client requirements, improving project efficiency and customer satisfaction (Z11).<br>I tried using generative AI to provide design ideas and help me find new creative directions (Z12). |
| A09 Task-specific Benefits | Accelerated draft generation<br>Intelligent information extraction<br>Creative inspiration | Generative AI helps me create drafts more quickly, significantly improving the efficiency of early-stage solution discussions (Z12).<br>Now I am finally freed from demand analysis—AI can extract key information from massive data and provide me with valuable suggestions (Z05).<br>I have tried using generative AI to offer design ideas and help me find new creative directions (Z14). |
| A10 Limited Applicability | Insufficient design standards and process compatibility<br>Dependence on human supervision | Although AI can help with some general design tasks, its application is limited in actual product design processes because it doesn't understand our product design team's standards, requirements, and workflows (Z18).<br>While AI may be helpful, I still must check and understand its generated designs or suggestions no matter what. So, I'm not sure whether it truly makes me more efficient (Z30). |
| A11 Ease of use | Learning curve adaptability<br>Optimized interaction experience<br>Intuitive interface | Once I became familiar with the AI tool interface, it became very convenient to use (Z15).<br>Conversational AI like GPT has enhanced my user experience and work efficiency (Z21).<br>Midjourney's interface is more intuitive than SD, making it easier for beginners to adopt (Z7). |

**Table 3.** *Cont.*

| Subcategory | Initial Category (Partial) | Original Sentence (Partial) |
| --- | --- | --- |
| A12 Design Goal Relevance | Alignment with design objectives<br>Effective goal achievement support<br>Innovation-enabling functionality | MJ and SD can handle almost all my design tasks, from sketches to refined renderings (Z08).<br>The automated design features of this AI tool perfectly meet my project needs for quickly generating graphics (Z29). |
| A13 Design Process Compatibility | Team collaboration compatibility<br>Seamless workflow integration | It can be integrated with team collaboration tools, allowing generated sketches to be directly shared with other team members for feedback, which improves collaboration efficiency (Z11).<br>Generative AI can directly integrate with my existing design software, such as Photoshop and Figma 2024 beta, enabling me to complete tasks without switching platforms (Z25). |
| A14 Intellectual Property Risks | Originality and copyright concerns<br>Design work attribution | There is currently little discussion about intellectual property issues related to AI-generated designs—could there be legal risks of infringement? (Z28)<br>Although the generated results are excellent, I am concerned that the design content may resemble existing works, leading to copyright issues (Z16). |
| A15 Data Privacy Risks | Potential privacy violations<br>Enterprise data security<br>Sensitive data protection | I don't know how this platform protects my design data (Z21).<br>Many models are not domestically developed, so there's no guarantee that they won't leak information (Z24; Z02). |
| A16 Peer Influence | Positive peer interaction<br>Supportive and collaborative learning community<br>Social circle acceptance | Interacting with my colleagues has positively influenced my perception of AI (Z29).<br>Many of my colleagues create amazing designs using SD, which motivated me to purchase an account (Z18).<br>I created a supportive and collaborative learning community on Xiaohongshu, which inspires me to actively engage in learning and using AI (Z21). |
| A17 Online Public Opinion | New media influence<br>Internet dissemination | Recently, I have seen AI-related topics trending in the news and on social media, and many people are discussing them online (Z26).<br>I would try an AI tool on Xiaohongshu or tiktok if it received positive reviews (Z13, Z15). |
| A18 Social Norms and Expectations | Social pressure for active participation<br>Inevitable industry trends | I feel a sense of social pressure to actively try AI and contribute (Z11).<br>I noticed that many job postings in professional communities require proficiency in AI tools, so I started learning to use them as well (Z08). |
| A19 Policy and Financial Support | Policy support<br>Product subscription support | The reimbursement process for these overseas product companies is very complicated, and sometimes they are not reimbursed at all (Z04).<br>The company has subscribed to popular AI products and encourages us to use them (Z13). |
| A20 Network and Hardware Support | Network reliability<br>Network accessibility<br>Hardware infrastructure support | Due to the company's network restrictions, using such tools is very inconvenient (Z12; Z14).<br>The enterprise network is highly reliable and easily accessible, ensuring that we can use AI tools smoothly (Z03).<br>I really appreciate the new equipment provided by the company, which minimizes hardware limitations (Z19). |

**Table 3.** *Cont.*

| Subcategory | Initial Category (Partial) | Original Sentence (Partial) |
| --- | --- | --- |
| A21 Education and Training | Regular training<br>Organizational promotion<br>Reliable and accessible technical support | Our company occasionally holds seminars on AI tools, which makes me consider integrating AI into my workflow (Z28).<br>I am very grateful for the technical training provided by the company, as it has enhanced my experience and minimized technical difficulties (Z19). |

#### 4.4.2. Axial Coding

Axial coding, the secondary coding process following open coding, involves analyzing and consolidating initial categories based on their logical relationships to form the main categories [84]. Based on the 21 subcategories identified through open coding, we established seven main categories that influence designers' Gen AI adoption intentions in CPD: personal innovativeness, AI technology anxiety, perceived usefulness, technology–task fit, perceived risk, social influence, and organizational support. These seven main categories encompassed the initial 21 preliminary categories (see Table 4). For instance, the main category "Individual Innovativeness" comprises three preliminary categories: proactivity and enthusiasm, exploratory motivation, and creative autonomy. These intrinsic psychological factors directly influence designers' intention to adopt Gen AI in CPD.

**Table 4.** Axial coding analysis results.

| Major Categories | Subcategory | Meaning |
| --- | --- | --- |
| Personal Innovativeness (PI) | A01 Proactivity and Enthusiasm<br>A02 Exploratory Motivation<br>A03 Creative Autonomy | Personal innovativeness refers to designers' tendency to demonstrate creative thinking and proactively explore new approaches when adopting Gen AI in CPD. |
| AI Technology Anxiety (ATA) | A04 Skill-related Anxiety<br>A05 Low Self-efficacy<br>A06 Career Development Anxiety | AI technology anxiety refers to designers' perceived anxiety when using Gen AI to complete design tasks in CPD. |
| Perceived Usefulness (PU) | A07 Work Efficiency Enhancement<br>A08 Design Quality Enhancement<br>A09 Task-specific Benefits<br>A10 Limited Applicability | Perceived usefulness refers to the degree to which designers believe that using Gen AI can enhance their work performance or design quality when completing design tasks in CPD. |
| Technology–Task Fit (TTF) | A11 Ease of use<br>A12 Design Goal Relevance<br>A13 Design Process Compatibility | Task–technology fit refers to the degree to which designers subjectively perceive that Gen AI's functionality, performance, and usability meet their design task requirements when using Gen AI to complete design tasks in CPD. |
| Perceived Risk (PR) | A14 Intellectual Property Risks<br>A15 Data Privacy Risks | Perceived risk refers to designers' subjective expectations of potential issues (such as copyright and data security) that may arise when using Gen AI to complete design tasks in CPD. |
| Social Influence (SI) | A16 Peer Influence<br>A17 Online Public Opinion<br>A18 Social Norms and Expectations | Social influence refers to the extent to which designers' adoption of Gen AI in CPD is influenced by the social environment, peers, and organizations. |
| Organizational Support (OS) | A19 Policy and Financial Support<br>A20 Network and Hardware Support<br>A21 Education and Training | Organizational support refers to the level of resources, environment, and policy support provided by enterprises or institutions to facilitate employees' adoption of new tools and technologies when using Gen AI to complete design tasks in CPD. |

#### 4.4.3. Selective Coding

Selective coding involves synthesizing the core category and refining the theoretical framework. In this study, we identified the core category as the factor influencing designers' intention to adopt Gen AI in CPD. Through an in-depth analysis, we identified seven main

categories and twenty-one subcategories that formed a structured conceptual model. This model provides a comprehensive framework of the factors influencing designers' Gen AI adoption, illustrating the relationships between the preliminary and main categories in a structured and interpretable manner (see Figure 2).

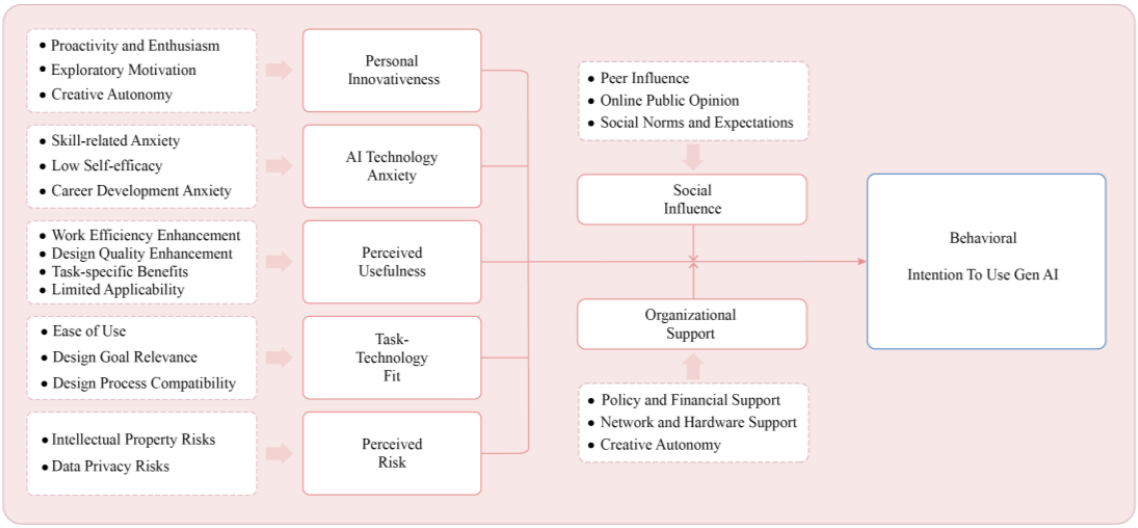



**Figure 2.** Factors influencing designers' adoption intention of Gen AI in CPD.

### *4.5. Theoretical Saturation*

The saturation test evaluates whether new concepts, categories, or relationships can be identified from additional samples [85]. Following the preliminary coding phase and based on Francis' recommendations, we conducted a saturation test using 15 reserved samples from the open coding phase [86]. The results validated the existing relational structure, indicating that no new concepts or categories emerged during the coding process. To further validate the coding analysis, we randomly selected and interviewed one design expert and two information system experts. A comparative analysis of their perspectives revealed no new categories or concepts, confirming that our coding results meet the criteria for theoretical saturation.

### *4.6. Reliability and Validity Test*

To validate the identified factors influencing designers' Gen AI adoption intentions, we assessed coding reliability with a team comprising one professor and two doctoral students with extensive research experience. We randomly selected interview samples for the preliminary analysis, with each researcher independently conducting coding according to the established objectives. The Holsti formula was applied to assess intercoder reliability [87]. The results exceeded 0.8, demonstrating a high coding consistency [88].

### *4.7. Findings of Study 1*

Based on our analysis, we developed a theoretical model explaining the factors that influence designers' Gen AI adoption intentions in CPD (see Figure 2). This model emphasizes that corporate designers' intentions to adopt Gen AI are shaped by both internal personal factors and external environmental factors. Internal factors refer to those existing within individuals, based on personal characteristics or cognition. For instance, personal innovativeness and AI technology anxiety reflect individual personality traits, whereas perceived usefulness, task–technology fit, and perceived risk reflect subjective evaluations of technological performance and risks. External factors refer to influences from the ex-ternal environment, where social influence represents the external impact of colleagues, leaders,

and social norms, while organizational support represents the driving force of enterprises or institutions in terms of resources, environment, and policies on individual behavior.

### 4.8. Conclusion of Study 1

In Study 1, we developed a comprehensive model examining both internal (personal innovativeness, AI technology anxiety, perceived usefulness, task–technology fit, and perceived risk) and external factors (social influence and organizational support). These findings not only partly confirm previous research on the factors influencing designers' Gen AI adoption beyond CPD [19,21,63,64] but also reveal that personal innovativeness and organizational support are important influencing factors in the CPD environment. Furthermore, our study expands on previous research by identifying seven main categories and 21 specific factors, offering deeper insights into the multidimensional nature of designers' Gen AI adoption intentions. This theoretical model aligns with Bandura's Social Cognitive Theory and Ryan and Deci's Self-Determination Theory, emphasizing the crucial role played by both internal personal and external environmental factors in shaping behaviors and outcomes [61,89]. Our findings not only reaffirm these foundational theories but also add to the growing body of knowledge by highlighting the importance of the interaction between internal factors and the external environment in facilitating designers' adoption of Gen AI in CPD.

## 5. Study 2: The Complexity of Designers' Gen AI Adoption Intention: A Configurational Analysis Based on fsQCA

### 5.1. Data Collection and Measurement

The questionnaire consisted of two main parts. The first part investigated designers' demographic information, including gender, age, professional experience, and experience with Gen AI. The second part employed a seven-point Likert scale to measure the seven antecedent variables and one outcome variable in the model. The measurement of variables in this study was derived from the coding results of grounded theory and built upon the existing literature to ensure the reliability and validity of the scales. After developing the initial questionnaire, feedback was solicited by two experts and modifications were made according to their suggestions (see Table 5).

This study adopted a two-stage data collection process to ensure sample continuity and representativeness. In the first stage, 215 questionnaires were distributed to interviewees from Study 1 and their colleagues. In the second stage, 162 additional questionnaires were distributed to product design professionals across various industries and regions in China, including product and industrial designers, UX/UI designers, user researchers, and product managers. To ensure that the respondents aligned with the research objectives, a screening question "Have you used Gen AI in your design work?", was implemented to exclude individuals without Gen AI experience. Data were collected online through the SurveyMars platform from 25 October 2024 to 9 November 2024. Of the 377 questionnaires distributed, 364 were returned and 13 were excluded because they did not meet the participation criteria. This study established several criteria to identify potentially problematic responses and ensure the reliability and completeness of the survey. First, questionnaires completed in less than 90 s were eliminated [90]. Additionally, samples showing obvious logical contradictions between the responses were excluded. Furthermore, questionnaires in which more than 80% of the responses were identical were deemed invalid. Responses displaying obvious patterns (such as sequential responses 1, 2, 3, 4, etc.) were also considered invalid [91]. After eliminating 37 inadequate responses, 327 valid responses were obtained. The demographic distribution of valid responses showed a relatively balanced gender ratio, with males accounting for 48.6% and females for 51.4%. Regarding the age

distribution, 28.9% of the respondents were 18–25 years old, 48.5% were 26–35 years old, and 22.6% were over 35 years old. Regarding work experience, the majority (43.1%) had more than three years of professional experience.

**Table 5.** Behavioral measurement scale.

| Constructs | Items | References |
| --- | --- | --- |
| Personal Innovativeness | PI1: If I heard about a new technology like Gen AI, I would look for ways to experiment with it<br>PI2: Among my peers, I am usually the first to try out new technologies like Gen AI.<br>PI3: I like to experiment with new technologies | [92] |
| AI Technology Anxiety | ATA1: I am concerned about becoming overly dependent on Gen AI<br>ATA2: I worry that Gen AI might threaten my job security<br>ATA3: I find it challenging to learn how to use Gen AI | [20,93] |
| Perceived Usefulness | PU1: Using LLMs would enhance my effectiveness on the job<br>PU2: Using Gen AI would improve the quality of work I do<br>PU3: Using Gen AI is helpful for generating creative inspiration | [94,95] |
| Technology–Task Fit | TTF1: It would be easy for me to become skillful at using Gen AI<br>TTF2: Using Gen AI is compatible with all aspects of my work<br>TTF3: I find Gen AI integrates well into my current workflow | [96] |
| Perceived Risk | PR1: I am concerned about personal privacy breaches when using Gen AI<br>PR2: I worry about copyright issues associated with Gen AI usage<br>PR3: I am concerned about the potential leakage of corporate design project information through Gen AI | [20,21] |
| Social Influence | SI1: My colleagues', supervisors', and friends' suggestions or behaviors influence my use of Gen AI<br>SI2: The intelligent development trends in the design field influence my use of Gen AI<br>SI3: The promotion and advocacy of Gen AI in the design field influence my usage | [97,98] |
| Organizational Support | OS1: My company provides network and hardware support for Gen AI usage<br>OS2: My company has policies and financial support for Gen AI adoption<br>OS3: My company offers training courses to help me learn Gen AI | [94,99] |
| Intention to Use | IU1: I am satisfied with Gen AI<br>IU2: I am willing to regularly use Gen AI to assist in my design work in the future<br>IU3: I am willing to recommend Gen AI to my colleagues | [98,100] |

### 5.2. Reliability and Validity

As shown in Table 6, Cronbach's alpha for all variables exceeded 0.8, indicating good reliability. The standardized factor loadings were above 0.5, and the composite reliability (CR) values exceeded 0.7, reflecting high internal consistency. All average variance extracted (AVE) values were above 0.5, confirming that the measurement items effectively explained their latent variables [101]. Therefore, the measurement model exhibits satisfactory convergent validity [102]. Table 7 further reveals that the square root of the AVE for each variable (on the diagonal) was greater than its correlations with other variables, confirming good discriminant validity [103].

**Table 6.** Results of reliability and validity tests.

| Constructs | Indicators | Loading | Cronbach α | CR | AVE |
|---|---|---|---|---|---|
| Personal Innovativeness | PI1 | 0.823 | 0.779 | 0.871 | 0.693 |
| | PI2 | 0.770 | | | |
| | PI3 | 0.865 | | | |
| AI Technology Anxiety | ATA1 | 0.846 | 0.845 | 0.911 | 0.774 |
| | ATA2 | 0.845 | | | |
| | ATA3 | 0.861 | | | |
| Perceived Usefulness | PU1 | 0.790 | 0.869 | 0.867 | 0.687 |
| | PU2 | 0.824 | | | |
| | PU3 | 0.789 | | | |
| Technology–Task Fit | TTF1 | 0.838 | 0.881 | 0.908 | 0.769 |
| | TTF2 | 0.856 | | | |
| | TTF3 | 0.861 | | | |
| Perceived Risk | PR1 | 0.855 | 0.841 | 0.897 | 0.744 |
| | PR2 | 0.835 | | | |
| | PR3 | 0.850 | | | |
| Social Influence | SI1 | 0.791 | 0.793 | 0.844 | 0.647 |
| | SI2 | 0.732 | | | |
| | SI3 | 0.794 | | | |
| Organizational Support | OS1 | 0.824 | 0.834 | 0.916 | 0.787 |
| | OS2 | 0.868 | | | |
| | OS3 | 0.872 | | | |
| Intention to Use | IU1 | 0.754 | 0.898 | 0.835 | 0.630 |
| | IU2 | 0.788 | | | |
| | IU3 | 0.743 | | | |

**Table 7.** Discriminant validity.

| | PI | ATA | PU | TTF | PR | SI | OS | IU |
|---|---|---|---|---|---|---|---|---|
| PI | 0.833 | | | | | | | |
| ATA | 0.486 | 0.880 | | | | | | |
| PU | 0.403 | 0.521 | 0.829 | | | | | |
| TTF | 0.446 | 0.545 | 0.426 | 0.877 | | | | |
| PR | 0.415 | 0.512 | 0.376 | 0.421 | 0.862 | | | |
| SI | 0.372 | 0.481 | 0.365 | 0.389 | 0.361 | 0.804 | | |
| OS | 0.309 | 0.387 | 0.271 | 0.316 | 0.293 | 0.341 | 0.887 | |
| IU | 0.411 | 0.462 | 0.388 | 0.421 | 0.396 | 0.374 | 0.341 | 0.794 |

### *5.3. Fuzzy Set Qualitative Comparative Analysis (fsQCA)*

#### 5.3.1. Calibration

In accordance with the fsQCA methodology requirements, data calibration was performed prior to conducting a fuzzy set qualitative comparative analysis. Following Ragin's calibration method, the 0.95, 0.5, and 0.05 quantiles of each causal variable were used as calibration thresholds for full membership, crossover points, and full non-membership, respectively [104]. These thresholds established fuzzy set calibration anchors for each variable, converting the questionnaire data into continuous membership scores within the interval [0, 1]. To minimize data exclusion in truth table construction owing to excessive values of 0.5, a constant of 0.001 was uniformly added to all calibrated values [105]. The variable anchor points are listed in Table 8.

**Table 8.** Calibration values.

| Conditions and Outcomes | Full Membership Threshold (95%) | Crossover (50%) | Full Non-Membership Threshold (5%) |
|---|---|---|---|
| PI | 6.000 | 4.667 | 1.667 |
| ATA | 5.667 | 4.000 | 2.000 |
| PU | 6.000 | 3.667 | 2.000 |
| TTF | 6.000 | 4.667 | 1.984 |
| PR | 6.000 | 3.667 | 1.653 |
| SI | 6.333 | 5.000 | 2.667 |
| OS | 5.667 | 3.000 | 1.667 |
| IU | 6.000 | 5.000 | 2.000 |

#### 5.3.2. Analysis of Necessary Conditions

Owing to the asymmetric assumptions of fsQCA, necessity analysis requires independent testing of the necessity of each causal condition [106]. Therefore, this study conducted a necessity analysis by independently testing each causal condition. A causal condition was considered necessary when the consistency score exceeded the threshold value of 0.90 [107]. The results revealed that the highest consistency score among all conditions was 0.751, indicating that no single condition was necessary to generate Gen AI adoption intentions (see Table 9).

**Table 9.** Analysis of necessary conditions for the conditions and outcomes.

| Antecedent Conditions | Strong Intention | | Weak Intention | |
|---|---|---|---|---|
| | Consistency | Coverage | Consistency | Coverage |
| PI | 0.706 | 0.710 | 0.631 | 0.583 |
| ~PI | 0.585 | 0.633 | 0.685 | 0.682 |
| ATA | 0.641 | 0.710 | 0.519 | 0.505 |
| ~ATA | 0.553 | 0.567 | 0.702 | 0.631 |
| PU | 0.704 | 0.751 | 0.523 | 0.490 |
| ~PU | 0.522 | 0.555 | 0.734 | 0.685 |
| TTF | 0.749 | 0.763 | 0.553 | 0.494 |
| ~TTF | 0.503 | 0.562 | 0.734 | 0.719 |
| PR | 0.685 | 0.734 | 0.540 | 0.508 |
| ~PR | 0.541 | 0.573 | 0.717 | 0.666 |
| SI | 0.710 | 0.676 | 0.677 | 0.565 |
| ~SI | 0.543 | 0.657 | 0.612 | 0.649 |
| OS | 0.597 | 0.640 | 0.665 | 0.627 |
| ~OS | 0.652 | 0.690 | 0.619 | 0.573 |

#### 5.3.3. Selection Criteria for fsQCA Indicators

Following data calibration, the next step involves creating a truth table as the foundation for the final solution and its various configurations. This truth table encompasses all possible causal combinations of the conditions. Given that this study involved 327 sample cases, exceeding the baseline of 150 cases, we set the frequency threshold to 3, consistency threshold to 0.9, and PRI consistency threshold to 0.7 to determine the final fsQCA solutions [108]. In solution identification, conditions appearing in both parsimonious and intermediate solutions are considered core conditions, whereas those appearing only in intermediate solutions are treated as peripheral conditions [109]. In terms of result notation, ● and ⊗ represent the presence and absence of core conditions, respectively, while • and $^{\otimes}$ denote the presence and absence of peripheral conditions, and blank spaces indicate that the condition may or may not be present. The consistency of each solution was reported, like significance testing, indicating the level of configuration required to produce the results. Additionally, coverage was provided for each solution, illustrating the degree to which

the combined configuration of the seven antecedents determined the Gen AI adoption intention in CPD [110].

### 5.4. Robustness Analysis

In qualitative comparative analysis, robustness testing of configurational paths is crucial due to the subjectivity in selecting thresholds and cutoff values, which may pose parameter-setting threats. Common methods for QCA robustness testing include adjusting calibration thresholds, case frequency thresholds, and consistency thresholds [69]. This study used the consistency threshold adjustment method, changing the original threshold from 0.9 to 0.8. The new truth table for high Gen AI designer adoption intention paths showed minimal differences from the original findings, confirming the robustness and reliability of the solution paths.

### 5.5. Findings and Discussion of Study 2

This study identified five configurational combinations associated with high adoption intention, revealing the strong inclination of designers to use Gen AI (see Table 10). These configurations demonstrated high consistency values above 0.9, indicating strong reliability. Configurations H1–H5 represent sufficient conditions for high adoption intention. Further classification of antecedent configurations with identical core conditions led to the identification of three driving patterns.

- Task demand-driven: In configuration H1 (PI*PU*TTF*PR; consistency = 0.924), the core conditions are PU and TTF, whereas high PR and PI function as peripheral conditions. Configuration H2 (~ATA*PU*TTF*SI; consistency = 0.908) also features PU and TTF as core conditions, with low ATA and high SI serving as peripheral conditions. Both configurations share high perceived usefulness and high task–technology fit as core conditions, indicating that designers' adoption intention toward Gen AI is primarily influenced by the combined effect of these two factors. This indicates that designers are more likely to adopt Gen AI when they perceive it as effective for their practical tasks and compatible with their design workflow. These findings align with previous research emphasizing the crucial roles played by perceived usefulness and task–technology fit in promoting technology adoption intention [21,63,92].
- Furthermore, in path H1, personal innovativeness emerges as a core condition and perceived risk exists as a peripheral condition, suggesting that highly innovative designers may adopt Gen AI when they meet their design task requirements, despite security and privacy concerns. This finding contradicts those of previous studies. Prior studies applying the perceived risk theory examined various risk dimensions (functional, psychological, and social) and established a negative correlation between perceived risk and behavioral intention [20,111,112]. However, as demonstrated in Study 1, we incorporated personal privacy and collective rights risks as the measurement dimensions. The results indicate that even when considering privacy, information security, and copyright issues, highly innovative designers maintain a positive attitude toward Gen AI and intend to implement it in CPD. This further reveals a "risk–technology benefit" trade-off mechanism in designers' technology adoption decisions within CPD contexts. The underlying reason for this mechanism may be that highly innovative individuals often exhibit a strong motivation for professional self-actualization and a high sense of self-efficacy [113,114]. These psychological traits make them more inclined to amplify the perceived value of technological innovation while downplaying potential privacy and security concerns [115]. As one designer stated, "Although AI poses potential security and privacy risks, its ability to enhance work performance makes mastering new technologies essential for career development, and I am confident in navigating

these challenges." (Z17, Senior Designer). Moreover, highly innovative individuals typically exhibit stronger risk-taking propensities [116,117]. This characteristic enables them to adopt a more positive attitude when weighing benefits against risks and to be willing to accept potential security and privacy risks due to significant innovation and performance improvements [118,119].

- In path H2, technology anxiety is absent as a core condition, while social influence exists as a peripheral condition, indicating that designers with low technology anxiety can enhance their Gen AI adoption intention when influenced by their peer groups. This is because lower technology anxiety enhances the consistency between peer-provided information and other perceived information [120].
- Organizational environment-driven: In configuration H3 (~ATA*PU*SI*OS; Consistency = 0.936), the core conditions are SI and OS, with low ATA and high PU as peripheral conditions. This demonstrates that when designers with low technology anxiety perceive Gen AI as useful, regardless of their personal innovativeness and security–privacy risks, a favorable external environment can stimulate high adoption intention. As one design department supervisor explained: "When promoting AI design tools, our company not only provided financial and training support, but also established a mechanism for designers to share experiences. I noticed that team members showed significantly increased interest in adopting new technology under such a supportive environment." (Z11, Design Supervisor). This pattern may stem from designers with low technology anxiety being more capable of perceiving and utilizing organizational support for Gen AI adoption [121]. Additionally, positive information from the organizational environment (such as peer influence and supportive policies) forms a positive interaction with designers' technological cognition, thereby reinforcing their adoption intentions [120]. This finding reveals the catalytic role of organizational environmental factors in technology adoption, which may be more prominent in the collectivist cultural context of Chinese enterprises [122].
- In configuration H4 (PI*~PU*SI*OS; consistency = 0.961), the core conditions are SI and OS, with low PU and high PI as peripheral conditions. This further highlights the importance of a favorable organizational environment: regardless of security–privacy risks or designers' technology anxiety, even when Gen AI does not directly demonstrate significant perceived usefulness, the combination of organizational support and social influence can still drive high adoption intention among designers. This may be because positive organizational perceptions and support effectively reduce individual sensitivity to risks and technology anxiety [123–125] while simultaneously shaping their perception of technology usefulness [126]. As one designer reflected, "Seeing colleagues using AI tools and improving efficiency, coupled with strong company support, naturally reduced my previous concerns about using AI" (Z30, Junior Designer). However, these findings were not unprecedented. Previous AI adoption studies based on UTAUT have emphasized the crucial role played by external environmental influence and support in shaping behaviors and outcomes [21,127,128]. Our research not only reaffirms these fundamental theories but also contributes to the growing body of knowledge by emphasizing the importance of creating a supportive environment in CPD to promote Gen AI adoption.
- Individual characteristic-driven: Interestingly, our findings reveal that combinations of designers' personal characteristics play a unique role in driving Gen AI adoption. In configuration H5 (PI*~ATA*TTF*~OS; consistency = 0.902), the core conditions are high PI and low ATA, with high TTF and low OS as peripheral conditions. This demonstrates that designers with high personal innovativeness and low technology anxiety exhibit strong adoption intentions when they perceive Gen AI as compatible

with design tasks, even in contexts of limited organizational support. The underlying mechanisms of this pattern can be attributed to several factors. First, individuals with high personal innovativeness demonstrate stronger autonomous learning capabilities and an exploratory spirit [129]. They tend to master new technologies through self-exploration, thus reducing their dependence on external environmental support [130]. As one product designer explained: "Although the company lacks systematic AI training, I am accustomed to actively exploring new technology applications independently, such as through online learning or exchanging experiences with designers from other companies" (Z09, Product Designer). This perspective is corroborated by Wu, who found that innovative tendencies can effectively reduce the psychological burden of technology usage by enhancing self-efficacy [131]. Moreover, low technology anxiety alleviates designers' concerns and apprehensions when using Gen AI, creating positive feedback loops [132,133]. Research indicates that lower technology anxiety positively moderates technology expectations, subsequently enhancing adoption intention [134,135]. Our study found that designers with low technology anxiety maintained more open attitudes and positive emotions, even without substantial external support.

- Finally, the rapidly evolving business environment amplifies the impact of personal characteristics. Unlike previous studies that have focused on stable scenarios, corporate designers must address the challenges of rapidly changing client demands and continuous innovation. This heightens the significance of personal traits in driving Gen AI adoption. As one product designer explained: "In the design field, innovation is a survival skill. Those willing to experiment with new technologies often find breakthroughs in their work" (Z09, Product Designer). Consequently, organizations with limited resources should prioritize designers who exhibit high innovativeness and low technology anxiety. These individuals can rapidly master new technologies through self-motivation, becoming key drivers in promoting Gen AI adoption. This finding extends the UTAUT model by highlighting the critical role of designers' personal characteristics in Gen AI adoption, particularly within the highly dynamic context of CPD.

**Table 10.** Configurations that lead to high adoption intention.

| Causal Conditions | High Adoption Intention | | | | |
|---|---|---|---|---|---|
| | **H1** | **H2** | **H3** | **H4** | **H5** |
| PI | ● | ⊗ | | ● | ● |
| ATA | | | ⊗ | | ⊗ |
| PU | ● | ● | ● | ⊗ | |
| TTF | ● | ● | | | ● |
| PR | ● | | | | |
| SI | | ● | ● | ● | |
| OS | | | ● | ● | ⊗ |
| Consistency | 0.924 | 0.908 | 0.936 | 0.961 | 0.902 |
| Raw coverage | 0.439 | 0.408 | 0.241 | 0.341 | 0.244 |
| Unique coverage | 0.064 | 0.114 | 0.039 | 0.031 | 0.057 |
| Overall consistency | | | 0.889 | | |
| Overall coverage | | | 0.721 | | |

Note. PI = personal innovativeness; ATA = AI technology anxiety; PU = perceived usefulness; TTF = technology–task fit; PRs = privacy risks; SI = social influence; OS = organizational support. ● = core casual condition (present). ● = peripheral casual condition (present). ⊗ = core casual condition (absent). ⊗ = peripheral casual condition (absent). Blank spaces indicate that a condition may be either present or absent.

In conclusion, configurations H1 and H2 emphasize the importance of PU and TTF combinations in promoting designers' Gen AI adoption intention. Configurations H3 and

H4 reveal that positive combinations of organizational environmental factors, such as social influence (SI) and organizational support (OS), can drive designers' Gen AI adoption. Additionally, configuration H5 suggests that Gen AI adoption may also be influenced by designers' personal characteristics.

## 6. Conclusions

This study employed grounded theory and fsQCA to systematically examine the key factors influencing designers' adoption intentions toward Gen AI in CPD and analyzed their interrelationships. Furthermore, it elucidated the adoption pathways and underlying mechanisms that influence designers' adoption of Gen AI. These findings highlight the following two main points. First, individual cognitive factors, including personal innovativeness, AI-related technological anxiety, perceived usefulness, technology–task fit, and perceived risk, along with organizational environmental factors such as social influence and organizational support, jointly constitute the key factors influencing designers' decisions to adopt Gen AI. This indicates that, in the CPD context, designers' adoption decisions are influenced not only by their intrinsic cognitive factors but also significantly shaped and guided by the organizational environment. Secondly, the fsQCA analysis indicates that no single antecedent condition is sufficient to determine designers' adoption of Gen AI. Instead, adoption is influenced by the combined effects of multiple factors, with interactions between different factors leading to distinct adoption pathways such as task demand-driven, organizational environment-driven, and individual characteristic-driven.

These findings underscore the importance of collaboration at both the individual and organizational levels in driving the adoption of Gen AI. Therefore, enterprises not only need to enhance designers' understanding of and adaptability to AI technology but also to optimize the organizational environment by providing a more supportive system for designers, thereby promoting the broader and more effective application of Gen AI in corporate product design.

### *6.1. Theoretical Implications*

First, this study integrates the context of CPD with designers' adoption intention of Gen AI, thereby broadening the research perspective. As a relatively new research topic, the adoption of Gen AI by designers in CPD has yet to receive widespread academic attention. Existing studies have predominantly focused on the adoption intentions of general users or explored designers' adoption behavior at the individual level. However, in the CPD context, designers' Gen AI adoption decisions are influenced not only by individual factors but also by a combination of multiple factors, including organizational environment. Unlike previous research, this study is the first to systematically examine designers' Gen AI adoption intention from a CPD perspective, enriching the research framework of new technology adoption in the design field.

Second, this study constructs an integrated model of factors influencing designers' Gen AI adoption intentions. Through grounded theory analysis, seven key influencing factors were identified, forming a relatively comprehensive theoretical framework. The findings not only partly confirm previous research on the factors influencing designers' Gen AI adoption beyond CPD but also reveal context-specific influencing factors (innovativeness and organizational support) in CPD.

Finally, although fsQCA has gained increasing attention in technology adoption research, particularly in studies on consumer robot adoption [27,136,137], it has not yet been applied to research on designers' Gen AI adoption intention. This study is the first to introduce the fsQCA method to uncover the complex interplay of multiple factors in designers' adoption of Gen AI. Unlike previous studies that focused on the net effect of

individual factors, this study identifies five distinct adoption pathways, demonstrating that designers' adoption intention is driven by specific configurations of multiple factors. This not only contributes methodological innovation to the study of technology adoption in the design field but also offers a valuable supplement to the net-effect thinking prevalent in prior research.

### 6.2. Practical Implications

Our findings provide important guidance for Gen AI implementation in the context of CPD. Based on the three identified driving patterns, we recommend that organizations adopt more targeted implementation strategies to enhance Gen AI adoption.

First, organizations should start with low-risk initiatives and prioritize investments in designers' AI knowledge. Our research reveals that designers' technological cognition and attitudes significantly influence their adoption intentions (configurations H1 and H2). Organizations should establish systematic knowledge development systems. Cooper and McCausland noted that effective knowledge training can reduce employee resistance to new technologies, enhance technological self-efficacy, and build trust [12]. Specifically, organizations should implement tiered training programs encompassing three levels: AI fundamentals, practical applications, and advanced innovation. In addition, organizations can establish innovation projects to encourage designers to explore new application scenarios.

Second, organizations should cultivate a positive adoption atmosphere to promote Gen AI applications. Our research finds that social influence plays a crucial role in multiple configurations (e.g., configurations H2 and H3), indicating that peer effects and organizational climate significantly influence designers' adoption decisions. Organizations should establish knowledge-sharing mechanisms through regular technical seminars and application case libraries to facilitate the exchange of experiences. Simultaneously, influential designers can be selected as opinion leaders to demonstrate and motivate team-wide adoption through their exemplary roles.

Third, organizations should develop comprehensive organizational support systems. The results from configurations H3 and H4 indicate that strong organizational environmental support significantly promotes Gen AI adoption. At the infrastructure level, organizations should invest in adequate hardware resources and software support to ensure the smooth operation of Gen AI tools. Regarding financial support, organizations need to establish dedicated budgets for infrastructure development, talent cultivation, and technical services. Continuous and stable financial support demonstrates organizational commitment and enhances adoption confidence. Furthermore, for technical support, organizations should establish professional support teams, implement rapid response mechanisms, and regularly update their technical guidelines. This comprehensive organizational support system plays a crucial role in promoting the sustained and deep application of Gen AI in CPD.

Finally, given limited organizational resources, companies need to adopt more targeted resource allocation strategies. Research indicates that designers with high innovativeness and low technology anxiety are more likely to adopt Gen AI. Therefore, in resource-constrained small- and medium-sized enterprises, priority should be given to designers' innovative practices, enabling them to accumulate experience and influence other designers. This strategy not only improves resource utilization efficiency but also creates positive demonstration effects.

### *6.3. Limitations and Future Research*

Several limitations of this study should be considered when interpreting the results and implications. First, our study sample primarily consisted of medium and large enterprises in China, which may limit the generalizability of our findings to other countries and small enterprises. Future research should extend this scope to include different national contexts and smaller firms. Second, our use of the grounded theory approach may introduce researcher subjectivity and potential interpretative bias. Although we have made conscious efforts to minimize subjectivity, we acknowledge that biases may still exist in the coding and analysis processes. Designers with strong technical expertise and artistic knowledge are better equipped to overcome the limitations of generative AI. This study does not consider the impact of designers' educational and knowledge backgrounds on the adoption of generative AI. Therefore, future research could explore additional factors to provide a more comprehensive understanding of this phenomenon.

Finally, Gen AI is evolving at a rapid pace, particularly with the emergence of Lora-based customized AI, which may influence key variables and impact the pathways identified in this study. As AI personalization and customization advance, the effect of perceived usefulness on adoption intention may diminish, while the role of task–technology fit could become more significant. Moreover, customized AI demands greater technical support and resource investment, further emphasizing the critical role played by organizational support in the AI adoption process. Additionally, advancements in Gen AI may introduce new research variables, such as designers' perceived control over AI and the alignment between AI-generated content and individual creative styles. Future research could explore these dynamics through longitudinal studies to track AIGC's evolution and user adaptation patterns.

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## Abbreviations

The following abbreviations are used in this manuscript:

| | |
|---|---|
| Gen AI | generative artificial intelligence |
| CPD | corporate product design |
| fsQCA | Fuzzy-set qualitative comparative analysis |

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